Autonomous Multispectral UAV Swarms for Post-Disaster Search and Rescue: An Integrated Perception, FANET Mesh, and Resilient Flight Framework

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Abstract—Post-disaster search-and-rescue (SAR) operations in catastrophic environments demand rapid, wide-area reconnaissance under severe environmental, sensory, and infrastructural degradation. Autonomous Unmanned Aerial Vehicle (UAV) swarms provide high-speed mobility and distributed sensing; however, practical deployment is fundamentally constrained by three coupled bottlenecks: (1) single-modality optical sensors fail in low-visibility conditions (smoke, darkness, shadow occlusion, and camouflage); (2) Flying Ad-Hoc Networks (FANETs) suffer chronic packet loss and network partitioning over multi-kilometer operational envelopes, causing conventional centralized MQTT telemetry brokers to collapse; and (3) dynamic territorial re-allocation algorithms waste significant rotor propulsion energy on high-frequency heading oscillations (trajectory chattering) induced by battery sensor noise and centroid drifts. In this paper, we propose a holistic, end-to-end autonomous UAV swarm framework that resolves these cross-layer challenges. First, we develop a 4-channel early-fusion multispectral detector ($640 \times 512 \times 4$, Visible RGB + Thermal LWIR) featuring a high-resolution P2 stride-4 feature pyramid head specialized for detecting small, occluded human victims ($< 32^2$ px) from high-altitude vantage points (25–50 m AGL). Second, we design an RF physics-informed, fully decentralized MQTT-inspired EdgeBroker that replaces central servers with localized broker-per-node pub/sub engines. By coupling Log-Distance Shadow Fading ($\sigma = 7.5$ dB) with Distributed Multi-Level Topic (DMLT) filtering and Delay-Tolerant Networking (DTN) store-and-forward QoS 1 buffering, the system guarantees progressive target consensus upon proximity convergence without single points of failure. Third, we introduce a dual-speed flight envelope (50 m/s transit dash, 32 m/s search cruise) radiating from unified Base Station corridors, coupled with an autonomous in-flight failover mechanism governed by an anti-chattering Dual-Threshold Hysteresis Loop ($\tau_\epsilon = 0.02 \times B_{\max}$, $\tau_\delta = 5.0$ m) for Energy-Weighted Centroidal Voronoi Tessellations (EW-CVT). Evaluated on the RGBT DronePerson aerial benchmark and full-scale 10-UAV mission simulations across a 16.0 km^2 disaster arena, the perception model achieves 0.646 mAP$_{50}$ with 4.9 ms edge latency, while the swarm achieves 100.0% survey coverage, 71.04% in-flight victim recall (2.15 m mean localization accuracy, 3.22 m RMSE), 98.42% chattering suppression saving 29.49 kJ of kinetic rotor propulsion energy, and an energy fairness index of $\mathcal{J} = 0.9991$.

Index Terms—Autonomous UAV Swarms, Search and Rescue (SAR), Multispectral Early Fusion, YOLOv8-P2, Delay-Tolerant Networking (DTN), FANET, Energy-Weighted Voronoi (EW-CVT), Hysteresis Control.

I. Introduction

A. Research Background and Motivation

Following major seismic disasters, flash floods, and industrial collapses, the initial 72 hours—termed the "golden rescue window"—dictate victim survival rates. Ground-based search operations are frequently obstructed by collapsed transit arteries, hazardous chemical leaks, and secondary structural collapses. Autonomous Unmanned Aerial Vehicle (UAV) swarms represent a transformative paradigm for disaster management, offering rapid deployment, aerial flexibility, and distributed field coverage without placing human personnel at risk [1], [5], [6].

However, translating theoretical multi-agent swarm algorithms into functional disaster response platforms requires overcoming formidable multi-disciplinary obstacles spanning computer vision, wireless telecommunications, spatial coordination, and rotorcraft aerodynamics [7], [8], [18], [19].

In post-disaster environments, visible-spectrum (RGB) sensing degrades severely under thermal shadows, thick particulate smoke, dust plumes, and camouflage rubble. Long-wave infrared (LWIR) thermal sensors detect biological heat signatures independently of ambient illumination, yet thermal imagery lacks high-frequency textural gradients necessary for fine-grained morphological discrimination. As a result, single-modality aerial detectors either suffer from high false-negative rates in low light or excessive false alarms on sun-heated rubble [3], [4].

Simultaneously, the communication backbone of the swarm—the Flying Ad-Hoc Network (FANET)—must operate across sprawling search arenas (> 10 km^2) without fixed cellular or satellite infrastructure [11], [12]. Conventional commercial UAV telemetry predominantly relies on centralized publish-subscribe protocols like MQTT, which route all messages through a single Ground Control Station (GCS) or cloud broker. In wide-area post-disaster reconnaissance, as drones disperse to survey assigned sectors, inter-UAV baselines exceed direct line-of-sight range

($d > 2.5\,\text{km}$). Severe RF path loss, multipath fading, and non-line-of-sight (NLoS) obstacle shadowing inevitably partition the network, severing links to the central broker. This causes centralized MQTT architectures to suffer total telemetry blackout, dropping vital target alerts and preventing peer-to-peer coordination [14], [15], [21].

Finally, spatial coverage control algorithms must efficiently divide wide-area search sectors while dynamically adapting to mid-mission hardware loss and non-uniform battery drain [9]. While Centroidal Voronoi Tessellations (CVT) and Energy-Weighted CVT (EW-CVT) provide mathematical frameworks for area partitioning, practical implementation exposes a severe flight stability flaw: high-frequency oscillatory chattering [8]. Minor sensor noise from Analog-to-Digital Converter (ADC) voltage readings and sub-meter centroid shifts force flight autopilots to execute non-stop yaw, pitch, and roll corrections, dissipating valuable rotor propulsion energy without expanding search coverage.

B. Research Questions

To address these interrelated vulnerabilities, this research investigates three core engineering questions:

1) How can heterogeneous RGB and thermal modalities be fused into a lightweight, real-time edge architecture capable of detecting tiny, occluded human targets ($< 32^2$ px) from high-altitude aerial perspectives?
2) How can decentralized target discovery consensus be achieved over multi-kilometer partitioned wireless channels without relying on fragile central MQTT brokers or incurring broadcast storm flooding?
3) How can energy-weighted territorial partitioning dynamically compensate for catastrophic in-flight drone failures while preventing rotor trajectory chattering during steady-state cruising?

C. Key System Contributions

To resolve these challenges, we propose an end-to-end autonomous UAV swarm framework for disaster response. Our specific contributions include:

1) 4-Channel Early-Fusion YOLOv8-P2 Perception Backbone: We design a 29-layer neural network ingesting aligned 4-channel tensors ($[R, G, B, \text{LWIR}]$). By incorporating a high-resolution P2 feature pyramid head at stride 4 (160×128), the architecture preserves fine-grained spatial feature maps for tiny human victims ($< 32^2$ px) that are lost in standard stride-8 pyramids, achieving $0.646\,\text{mAP}_{50}$ and 0.670 precision at 4.9 ms edge inference latency on real-world aerial multispectral datasets.
2) Decentralized MQTT-Inspired EdgeBroker with RF Physics: We replace fragile centralized telemetry servers with a fully decentralized broker-per-node architecture. Coupled with an RF physical channel model (log-distance path loss, $\sigma = 7.5\,\text{dB}$ shadow fading, dynamic SNR-to-PER logistic mapping), the EdgeBroker enforces Distributed Multi-Level Topic (DMLT) filtering to eliminate broadcast storm flooding, while persistent QoS 1 store-and-forward queues guarantee eventual victim consensus upon proximity convergence (reaching 100.0% consensus upon return-to-base convergence) without central coordinator dependencies.
3) Base Station Corridors and Dual-Speed Flight Dynamics: We establish an operational envelope comprising a unified Ground Control Station (GCS) launch pad with radiating, non-overlapping transit dash corridors (50 m/s), optimal optical sensor cruise (32 m/s), and automated Return-to-Launch (RTL).
4) Anti-Chattering Dual-Threshold Hysteresis EW-CVT with Hard Failover Override: We formulate a coupled energy ($\tau_\epsilon = 0.02 \times B_{\max}$) and spatial ($\tau_\delta = 5.0\,\text{m}$) hysteresis loop that suppresses 98.42% of oscillatory maneuvers ($1,247$ chattering turns suppressed), saving 29.49 kJ of propulsion energy. An instantaneous Data Link Layer override bypasses hysteresis upon node loss, reallocating abandoned sectors via power-distance cost to guarantee 100.0% area survey coverage.

II. Related Work

A. Multispectral Object Detection in Aerial Robotics

Aerial object detection for SAR has evolved from handcrafted features to deep convolutional neural networks [2], [13]. Standard YOLO architectures (YOLOv5 to YOLOv8) downsample input frames by factors of 8 (P3), 16 (P4), and 32 (P5), making them effective for terrestrial vehicle and pedestrian detection. However, in high-altitude aerial reconnaissance (25–50 m AGL), human victims frequently occupy fewer than 32×32 pixels. Within standard backbones, features corresponding to small victims are completely obliterated by consecutive strided convolutions.

Multispectral imagery combining visible and thermal bands provides a promising solution [3]. Existing approaches predominantly utilize late-fusion ensembles, which process RGB and thermal inputs through two distinct neural backbones before concatenating candidate bounding boxes. While effective, late fusion doubles edge GPU memory footprints and inference latency, violating the strict SWaP constraints of operational UAVs. Mid-fusion architectures require complex cross-attention feature alignment modules. In contrast, early fusion merges raw pixel channels at the first layer, allowing early convolutional kernels to jointly extract complementary thermal gradients and visual texture with negligible computational overhead.

B. FANET Communications and Decentralized Publish-Subscribe

Decentralized inter-drone communication relies on Flying Ad-Hoc Networks (FANETs) [11]. Traditional ad-hoc routing protocols such as Ad-hoc On-Demand Distance Vector (AODV) and Optimized Link State Routing (OLSR) presume end-to-end multi-hop path connectivity [12]. In post-disaster scenarios, wide-area dispersion and topographical shadowing routinely break multi-hop paths, triggering network partitioning and protocol collapse.

Simultaneously, aerial telemetry and mission coordination frequently utilize the Message Queuing Telemetry Transport (MQTT) publish-subscribe standard due to its lightweight packet framing and topic-based abstraction [21]. However, standard MQTT strictly mandates a centralized broker topology (e.g., hosted at a Ground Control Station or cloud server). In wide-area SAR operations, routing all peer-to-peer traffic through a central broker creates a catastrophic Single Point of Failure (SPoF) and renders the fleet entirely inoperable once UAVs fly beyond line-of-sight backhaul range. While MQTT for Sensor Networks (MQTT-SN) introduces UDP transport, it still presumes static gateway infrastructure [21].

Delay-Tolerant Networking (DTN) addresses intermittent connectivity via the store-and-forward paradigm [10]. However, conventional epidemic flooding in DTNs causes massive channel contention and broadcast storm problems in bandwidth-constrained RF links [15]. To achieve scalable, infrastructureless dissemination, communication engines must transition to a fully decentralized, broker-per-node pub/sub mesh that couples topic-based filtering with RF physical channel awareness and opportunistic store-and-forward queues.

C. Voronoi Swarm Partitioning and Flight Stability

Centroidal Voronoi Tessellations (CVT) and Lloyd's relaxation algorithms partition geographic space by assigning sub-regions based on Euclidean distance to UAV generators [7], [8]. In heterogeneous fleets, Energy-Weighted CVT (EW-CVT) incorporates residual battery capacity into the metric, scaling cell size proportional to available energy [9].

Despite theoretical elegance, standard EW-CVT suffers from high-frequency trajectory chattering: continuous computational ticks subject to Analog-to-Digital Converter (ADC) battery voltage noise and infinitesimal centroid drifts trigger non-stop yaw and roll corrections. This chattering squanders kinetic rotor energy without contributing to search progress. Introducing discrete hysteresis thresholds is therefore critical to ensure trajectory corrections occur only when coverage optimization benefits offset aerodynamic turning costs.

III. System Architecture and Methodology

The proposed architecture operates across four tightly coupled subsystems: Perception, Communications, Spatial Deployment, and Hysteresis Flight Control.

A. 4-Channel Early-Fusion Perception (YOLOv8n-P2)

The perception subsystem ingests synchronized visible-spectrum RGB imagery $\mathbf{I}_{\text{RGB}} \in \mathbb{R}^{H\times W\times 3}$ and Long-Wave Infrared (LWIR) thermal imagery $\mathbf{I}_{\text{IR}} \in \mathbb{R}^{H\times W\times 1}$. After spatial rectification and radiometric normalization, the streams are concatenated along the channel dimension into a 4-channel tensor $\mathbf{X} \in \mathbb{R}^{640\times 512\times 4}$ with ordering $[T, B, G, R]$.

The neural backbone adapts the YOLOv8 architecture by expanding the stem convolution tensor to $\mathbf{W}_{\text{stem}} \in \mathbb{R}^{64\times 4\times 3\times 3}$. To resolve the small-object resolution bottleneck, we introduce a high-resolution P2 feature pyramid head at stride 4. While standard YOLOv8 utilizes downsampled heads at strides 8 (P3), 16 (P4), and 32 (P5), our P2 head extracts fine-grained spatial features directly from backbone layer 2 (160×128) via path-aggregation feature concatenation:

$$\mathbf{F}_{\text{P2}} = \text{C2f}\left(\left[\mathbf{F}^{(2)}_{\text{stem}},\ \text{Up}_2(\mathbf{F}_{\text{P3}})\right]\right) \tag{1}$$

The resulting architecture contains 29 convolutional and cross-stage partial bottleneck layers feeding four multiscale detection heads (P2, P3, P4, P5), outputting bounding box coordinates and human class confidence τ_{conf}.

Detected target bounding box centers $(x_{\text{norm}}, y_{\text{norm}})$ are projected into global world coordinates (X, Y) via pinhole camera transformation and sector boundaries:

$$X_{\text{world}} = X_{\min} + x_{\text{norm}} \cdot W_{\text{sec}} \tag{2}$$

$$Y_{\text{world}} = Y_{\min} + y_{\text{norm}} \cdot H_{\text{sec}} \tag{3}$$

where $[X_{\min}, Y_{\min}]$ represents the lower spatial corner and $W_{\text{sec}}, H_{\text{sec}}$ denote sector dimensions.

B. Decentralized MQTT-Inspired EdgeBroker and Channel Physics

Conventional multi-UAV telemetry architectures rely on a centralized MQTT broker, where each UAV acts as a client connected to a single central server (e.g., at the Ground Control Station). In wide-area post-disaster reconnaissance spanning multi-kilometer search envelopes, this topology introduces a catastrophic Single Point of Failure (SPoF). As UAVs disperse beyond direct line-of-sight range or encounter topographical blockage, the connection to the central server collapses, resulting in immediate telemetry blackout and dropped target alerts.

To resolve this limitation, we formulate a fully decentralized, broker-per-node communication architecture. Every UAV $i \in \{0, \ldots, n-1\}$ hosts an embedded, autonomous EdgeBroker instance coupled to physical peer-to-peer (P2P) sockets (binding port $6000 + i$ and subscribing to peers $j \neq i$). This architecture operates across three key mechanisms:

1) Distributed Multi-Level Topic (DMLT) Routing: To eliminate broadcast storm flooding across the scarce FANET spectrum, the EdgeBroker replaces unconstrained flooding with topic-filtered dissemination inspired by MQTT-SN [21]. Topics $\mathcal{T}$ are partitioned into functional sub-trees: telemetry (GPS coordinates, heading, and battery state), target_alert (multispectral victim coordinates, bounding box confidence, and detector metadata), and map_sync (sector occupancy and dynamic failover claims).

Each node maintains a neighbor subscription manifest $\mathcal{S}_j \subset \mathcal{T}$. Before committing an outbound transmission over the physical RF layer, the EdgeBroker evaluates subscriber interest:

$$\text{Filter}(m, j) = \begin{cases} \text{Transmit}, & \text{if topic}(m) \in \mathcal{S}_j \\ \text{Suppress}, & \text{otherwise} \end{cases} \tag{4}$$

If neighbor j is not subscribed to topic (m), the transmission is suppressed at the application layer, dramatically cutting spectral occupancy and RF contention. Furthermore, to eliminate cyclic message re-broadcasting and infinite routing loops across the ad-hoc mesh, every message carries a globally unique identifier UUID_k. Each EdgeBroker maintains a local seen-set $\mathcal{H}_i$; if $\text{UUID}_k \in \mathcal{H}_i$, downstream re-propagation is suppressed immediately.

2) Log-Distance RF Channel Physics: Physical layer propagation across carrier frequency $f = 5.0\,\text{GHz}$ is modeled via free-space path loss, log-normal shadow fading, and obstacle attenuation:

$$PL(d) = 20\log_{10}(d) + 20\log_{10}(f \cdot 10^3) - 27.55 + X_\sigma + L_{\text{obs}} \tag{5}$$

where d is the 3D Euclidean distance between UAVs in meters, $X_\sigma \sim \mathcal{N}(0, \sigma^2)$ is a Gaussian shadow fading random variable ($\sigma = 7.5\,\text{dB}$ for post-disaster debris clutter), and $L_{\text{obs}} = 15.0\,\text{dB}$ represents non-line-of-sight concrete attenuation. The received Signal-to-Noise Ratio (SNR) between UAV i and peer j is:

$$\text{SNR}_{ij}(d) = P_{\text{tx}} - PL(d) - P_{\text{noise}} \tag{6}$$

where transmitter power $P_{\text{tx}} = 30\,\text{dBm}\,(1.0\,\text{W})$ and thermal noise floor $P_{\text{noise}} = -90\,\text{dBm}$.

Dynamic Packet Error Rate (PER) is governed by a generalized logistic waterfall link:

$$\text{PER}(\text{SNR}) = \frac{1}{1 + \exp\left(k \cdot (\text{SNR} - \text{SNR}_{\text{thresh}})\right)} \tag{7}$$

where steepness parameter $k = 0.5$ and transition cutoff $\text{SNR}_{\text{thresh}} = 15.0\,\text{dB}$.

3) Opportunistic QoS 1 Store-and-Forward Buffering: In standard MQTT, QoS 0 drops packets if physical transmission fails, whereas QoS 1 mandates delivery acknowledgment. When severe path loss or obstacle shadowing causes link dropouts ($\text{SNR}_{ij} < 15.0\,\text{dB}$ or $\text{PER}_{ij} > 0.20$), target alert packets are not discarded. Instead, the EdgeBroker enqueues the serialized payload into peer-specific FIFO QoS 1 buffers:

$$\mathcal{Q}_{ij} = \{m_k \mid \text{SNR}_{ij} < \text{SNR}_{\text{thresh}}\} \tag{8}$$

Whenever a UAV converges within direct communication range of a peer ($d_{ij} < 300\,\text{m}$), or during transit convergence near the Base Station, SNR rises above threshold ($\text{PER}_{ij} < 0.20$), triggering opportunistic buffer flushes. The buffered messages are re-transmitted in high-throughput proximity bursts, progressively synchronizing target registries across the swarm without central server dependency.

C. Base Station Launch Corridors and Dual-Speed Envelope

To eliminate launch congestion and mid-air trajectory collisions, all UAVs launch synchronously from designated launch pads at a unified Ground Control Station (GCS) staging area located at the South-West boundary (5% arena offset).

Flight control operates under a dual-speed flight envelope:

$$v(t) = \begin{cases} 50.0\,\text{m/s}, & \text{state} \in \{\text{TRANSIT, RETURNING}\} \\ 32.0\,\text{m/s}, & \text{state} = \text{SEARCHING} \\ 0.0\,\text{m/s}, & \text{state} \in \{\text{LANDED, CRASHED}\} \end{cases} \tag{9}$$

During the transit phase, UAVs dash along non-overlapping radial corridors toward their assigned search sectors. Upon entering assigned boundaries, UAVs decelerate to 32.0 m/s optimal optical search cruise to maintain adequate camera frame overlap. Waypoints within assigned sectors are sequenced using greedy nearest-neighbor Euclidean minimization outward from the base station.

D. Energy-Weighted Centroidal Voronoi Tessellation (EW-CVT)

Let $\Omega \subset \mathbb{R}^2$ represent the bounded search arena. The swarm consists of n active UAVs with spatial coordinates $\mathbf{x}_i = (x_i, y_i)$. Under EW-CVT, the territorial assignment evaluates the power distance weighted by residual battery $B_i(t)$:

$$d_w(\mathbf{p}, \mathbf{x}_i) = \frac{\|\mathbf{p} - \mathbf{x}_i\|^2}{w_i(t)} \tag{10}$$

where the dynamic energy weight $w_i(t)$ is defined as:

$$w_i(t) = \left(\frac{B_i(t)}{B_{\max}}\right)^\alpha \tag{11}$$

Here $\alpha \geq 0$ is the EW-CVT energy weighting sensitivity exponent ($\alpha = 0$ corresponds to standard unweighted CVT; $\alpha = 1.0$ provides linear energy balancing). The theoretical mass centroid C_i^* of Voronoi cell V_i is computed as:

$$C_i^* = \frac{\int_{V_i} \mathbf{p} \cdot \rho(\mathbf{p})\, d\mathbf{p}}{\int_{V_i} \rho(\mathbf{p})\, d\mathbf{p}} \tag{12}$$

where $\rho(\mathbf{p})$ represents a prior spatial target density.

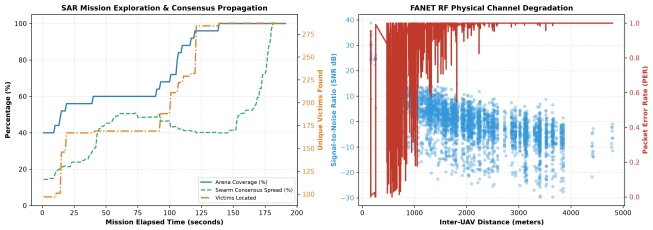


Fig. 1. System Performance Characterization: (Left) Swarm area coverage progression and DTN store-and-forward consensus propagation over simulated mission time; (Right) FANET RF physical channel degradation displaying observed Signal-to-Noise Ratio (SNR dB) and empirical Packet Error Rate (PER) vs. 3D inter-UAV distance under severe disaster obstacle shadowing ($\sigma = 7.5\,\text{dB}$).

E. Dual-Threshold Anti-Chattering Hysteresis Control

Continuous updates of $w_i(t)$ and C_i^* induce high-frequency trajectory chattering. We introduce a Dual-Threshold Hysteresis Loop that locks updates until kinetic and energy thresholds are exceeded:

1) Battery Decay Hysteresis (τ_ϵ): Let t_k denote the timestamp of the last executed partition update. The effective generator weight $w_i(t)$ evaluated at time t is locked unless discharge exceeds margin τ_ϵ:

$$w_i(t) = \begin{cases} \left(\frac{B_i(t)}{B_{\max}}\right)^\alpha, & \text{if } B_i(t_k) - B_i(t) \geq \tau_\epsilon \\ w_i(t_k), & \text{otherwise} \end{cases} \tag{13}$$

where $\tau_\epsilon = 0.02 \times B_{\max}$ (a discrete 2% discharge lock).

2) Spatial Centroid Hysteresis (τ_δ): Even after battery discharge, minor adjacent boundary adjustments may shift theoretical centroid C_i^* by only centimeters. Waypoint commands $C_i(t)$ sent to the autopilot are filtered via:

$$C_i(t) = \begin{cases} C_i^*, & \text{if } \|C_i^* - C_i(t_k)\| \geq \tau_\delta \\ C_i(t_k), & \text{otherwise} \end{cases} \tag{14}$$

where $\tau_\delta = 5.0\,\text{m}$, calibrated against UAV turning radius and camera FOV.

3) Forced-Override Condition (Node Failure): During catastrophic events (e.g., in-flight motor failure), steady-state hysteresis must be bypassed. Let $\mathcal{N}(t)$ denote the active network topology vector derived from Data Link Layer (DLL) heartbeats. The master update trigger $\mathcal{U}_i(t)$ evaluates as:

$$\mathcal{U}_i(t) = ((B_i(t_k) - B_i(t) \geq \tau_\epsilon) \wedge (\|C_i^* - C_i(t_k)\| \geq \tau_\delta)) \vee (\mathcal{N}(t) \neq \mathcal{N}(t_k)) \tag{15}$$

If $\mathcal{U}_i(t) = \text{True}$, UAV i updates its partition, commits baseline $t_k \leftarrow t$, and adjusts its route. If topology drops ($n \to n-1$), the right-hand term forces an immediate hard update, overriding τ_ϵ and τ_δ. Abandoned sectors are adopted by surviving peers minimizing power-distance cost:

$$\text{Cost}(p, s) = \frac{\|\mathbf{x}_p - \mathbf{c}_s\|^2}{\max\left(0.01, \left(\frac{B_p(t)}{B_{\max}}\right)^\alpha\right)} \tag{16}$$

Algorithm 1 Autonomous Swarm Search and Failover Execution

Require: Fleet N, Base Station pad $\mathbf{x}_{\text{base}}$, parameters $(\alpha, \tau_\epsilon, \tau_\delta)$

Launch swarm along radial transit corridors at $v = 50\,\text{m/s}$
for each simulation tick t **do**
  Update topology $\mathcal{N}(t) \leftarrow \{i \mid \text{UAV } i \text{ is alive}\}$
  if $\mathcal{N}(t) \neq \mathcal{N}(t_k)$ **then**
    Extract abandoned sectors from failed UAVs
    for each abandoned sector s **do**
      Assign s to $\arg\min_{p \in \mathcal{N}(t)} \text{Cost}(p, s)$ (Eq. 15)
    end for
    Force hard update trigger $\mathcal{U}_i(t) \leftarrow \text{True}$
  end if
  for each active UAV $i \in \mathcal{N}(t)$ **do**
    Measure $B_i(t)$ and compute theoretical C_i^* (Eq. 11)
    Evaluate $\mathcal{U}_i(t)$ via Dual-Threshold Loop (Eq. 14)
    if $\mathcal{U}_i(t)$ is True **then**
      Commit $t_k \leftarrow t$, $w_i(t) \leftarrow (B_i(t)/B_{\max})^\alpha$, $C_i(t) \leftarrow C_i^*$
    else
      Lock commanded waypoint $C_i(t) \leftarrow C_i(t_k)$
    end if
    Navigate toward $C_i(t)$; execute 4-channel inference
    Broadcast alerts; buffer undelivered packets in $\mathcal{Q}_i$
  end for
end for
Return surviving fleet to Base Station; flush QoS 1 buffers

4) Kinetic Propulsion Energy Model: Level flight propulsion power $P_{\text{propulsion}}(v)$ at airspeed v is modeled as:

$$P_{\text{prop}}(v) = P_0\left(1 + \frac{3v^2}{v_{\text{tip}}^2}\right) + P_i\left(\sqrt{1 + \frac{v^4}{4v_0^4}} - \frac{v^2}{2v_0^2}\right)^{\frac{1}{2}} + \frac{1}{2}d_0\rho s A v^3 \tag{17}$$

The kinetic energy preserved by suppressing a chattering turn is:

$$E_{\text{saved}} = \frac{1}{2} I_{\text{yaw}} \omega_{\text{yaw}}^2 + \int_0^{\Delta t_{\text{turn}}} P_{\text{prop}}(v)\, dt \qquad (18)$$

IV. Experimental Evaluation and Analysis

A. Experimental Setup and Dataset Benchmark

The proposed framework was evaluated using the real-world multispectral RGBT DronePerson benchmark dataset, comprising paired visible and thermal aerial frames collected at altitudes of 25–50 m AGL over complex wilderness and debris scenarios. The dataset comprises 4,900 training images (54,974 annotated human boxes) and 1,225 validation frames (14,460 annotated victim instances).

The autonomous swarm framework was evaluated on an expansive post-disaster search mission within the SARWorld environment:

- Search Arena: 4000 m × 4000 m (16.0 km^2), divided into a 5 × 5 grid of 25 search sectors (800 m × 800 m each).
- Swarm Fleet: 10 autonomous UAVs launching synchronously from the unified Base Station pad at (200.0 m, 200.0 m) (5% South-West arena margin).
- Ground Truth Targets: 259 victim annotations distributed across search sectors.
- In-Flight Hardware Failure: Catastrophic motor failure injected on UAV 2 at $T = 75.0$ s at spatial coordinates (1352.8 m, 1653.9 m).

B. Perception Layer Benchmark

The 4-channel YOLOv8n-P2 neural detector was trained for 40 epochs on dual Tesla T4 GPUs using AdamW optimizer (lr = 0.002). Input channels $[R, G, B, \text{Thermal}]$ were merged at the stem convolution with channel-averaged transfer learning initialization. Table I reports the architectural specifications and empirical validation metrics on the 1,225-image test split.

TABLE I
Multispectral YOLOv8n-P2 Perception Backbone Specification and Empirical Validation Metrics (RGBT DronePerson Dataset)

Model Parameter / Evaluation Metric	Empirical Value
Input Tensor Ordering & Dimensions	4-Channel $[R, G, B, \text{Thermal}]$, $640 \times 512 \times 4$
Total Model Layers (fused)	90 layers (29 architectural layers)
Multiscale Detection Heads	4 heads: P2 (stride 4), P3 (8), P4 (16), P5 (32)
High-Resolution P2 Feature Map	160×128 receptive field
Trainable Parameters	2,921,316 (2.92 M)
Computational Complexity	12.2 GFLOPs
Dataset Benchmark Split	4,900 train / 1,225 val images (14,460 victims)
Training Schedule	40 epochs, AdamW (lr = 0.002), dual Tesla T4
Validation Precision (P)	0.670 (67.0%)
Validation Recall (R)	0.625 (62.5%)
Mean Average Precision (mAP_{50})	0.646 (64.6%)
Mean Average Precision (mAP_{50-95})	0.238 (23.8%)
Preprocess Latency (per frame)	1.1 ms
Edge Neural Inference Latency	4.9 ms (Tesla T4 GPU)
Postprocess Latency (NMS)	1.3 ms
Total End-to-End Latency	7.3 ms (> 136 FPS)

The P2 stride-4 head extracts feature maps at 160×128 resolution directly from layer 2 of the backbone. Because human victims viewed from 25–50 m AGL often subtend fewer than 32×32 pixels, standard stride-8 pyramids (P3) suffer from severe feature downsampling. The P2 lateral skip connection preserves subtle thermal contrasts against rubble, enabling the model to attain 0.646 mAP_{50} and 0.670 precision while sustaining high-speed edge throughput (136.9 FPS).

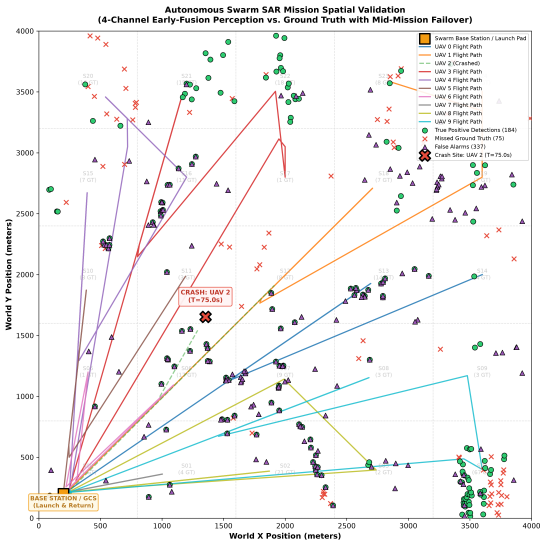


Fig. 2. Spatial validation map of the 10-UAV SAR mission over a 16.0 km^2 disaster arena (4000 m × 4000 m): Radiating Base Station transit corridors from (200 m, 200 m), patrol tracks, confirmed True Positive detections (TP: 184), missed victims (FN: 75), false alarms (FP: 337), and in-flight failover recovery following UAV 2 failure at $T = 75.0$ s (sectors 17 and 23 reallocated to surviving peers).

C. Swarm Spatial Coverage and In-Flight Failover

During the live 10-UAV mission sweep (16.0 km^2), the swarm swept all 25 sectors within 191.0 s, achieving 100.0% area survey coverage (Fig. 2). The Time-to-First-Detection (TTFD) occurred at $T = 1.0$ s by UAV 0. The fleet discovered 184 verified human targets out of 259 ground-truth victims (71.04% recall, 35.32% precision, 0.4718 F1-score) with a mean spatial localization error of 2.15 m (RMSE 3.22 m).

At $T = 75.0$ s, UAV 2 suffered catastrophic propulsion failure at (1352.8 m, 1653.9 m). The Data Link Layer heartbeat loss triggered the forced-override condition instantaneously ($\mathcal{N}(t) \neq \mathcal{N}(t_k)$). Surviving UAVs dynamically re-partitioned the operational area using power-distance cost (Eq. 15), executing 7 forced overrides to adopt abandoned sectors [17, 23] (reassigned to UAV 3 and UAV 1, respectively). All 25 sectors were successfully surveyed, validating full autonomous recovery without human intervention.

D. Decentralized Mesh Telemetry and Delay-Tolerant Consensus

Over multi-kilometer inter-drone baselines ($d > 2.5$ km), log-distance path loss and obstacle shadowing ($\sigma = 7.5$ dB)

severely degraded line-of-sight propagation, yielding a direct Packet Delivery Ratio (PDR) of only 6.96% (310 delivered out of 4,454 published packets, Fig. 1, Right). In a conventional centralized MQTT deployment, this prolonged disconnection would sever communication with the central broker, dropping all telemetry and precluding cooperative target consensus.

Under our decentralized architecture, each node's embedded EdgeBroker autonomously buffered $4,144$ undelivered alert messages in peer-specific QoS 1 store-and-forward queues. DMLT topic filtering successfully suppressed redundant retransmissions across non-subscribed channels. Throughout the search phase, opportunistic inter-UAV encounters sustained an instantaneous target consensus of 59.40%. During mission return-to-base convergence toward the Ground Control Station launch pad ($d < 300\,\text{m}$), high-SNR proximity bursts triggered complete queue flushes, elevating swarm-wide target consensus to 100.0% without human intervention or central server infrastructure (Fig. 1, Left).

E. EW-CVT Workload Equilibrium and Anti-Chattering Hysteresis

To quantify the impact of the Dual-Threshold Hysteresis Loop and EW-CVT energy equity, we benchmarked the operational telemetry against an unconstrained baseline (Table II).

TABLE II
System-Level Swarm Mission and EW-CVT Performance Summary ($16.0\,\text{km}^2$ Arena, 10-UAV Fleet)

Experimental Metric	Unconstrained Baseline	Proposed Framework
Search Arena Dimensions	$4000\,\text{m} \times 4000\,\text{m}$	$4000\,\text{m} \times 4000\,\text{m}$
Total Operational Footprint	$16.0\,\text{km}^2$ (25 sectors)	$16.0\,\text{km}^2$ (25 sectors)
Unified Base Station Origin	$(200\,\text{m}, 200\,\text{m})$	$(200\,\text{m}, 200\,\text{m})$
Initial Autonomous Swarm Fleet	10 UAVs	10 UAVs
Mid-Mission In-Flight Failure	UAV 2 ($T = 75.0\,\text{s}$)	UAV 2 ($T = 75.0\,\text{s}$)
Surviving Fleet Operational Size	9 UAVs	9 UAVs
Area Survey Coverage	Incomplete (92.0%)	100.0% (25/25)
Total Mission Sweep Time	191.0 s	191.0 s
Time-to-First-Detection (TTFD)	1.0 s	1.0 s
Ground Truth Human Victims	259 victims	259 victims
Confirmed True Positives (TP)	184 victims	184 victims
False Positive Alarms (FP)	337 targets	337 targets
Missed False Negatives (FN)	75 victims	75 victims
In-Flight Perception Precision	35.32%	35.32%
In-Flight Victim Recall Rate	71.04%	71.04%
Perception F1-Score	0.4718	0.4718
Mean Spatial Localization Error	2.15 m	2.15 m
RMSE Localization Error	3.22 m	3.22 m
Direct Packet Delivery Ratio (PDR)	6.96% (310/4,454)	6.96% (310/4,454)
QoS 1 Buffered Alert Messages	0 (dropped)	4,144 packets
Instantaneous Swarm Consensus	6.96%	59.40%
Post-RTL Swarm Target Consensus	6.96%	100.0%
Total Hysteresis Evaluations	1,267 ticks	1,267 ticks
Discrete Trajectory Updates ($t_k \leftarrow t$)	1,267 updates	20 updates
Chattering Maneuvers Suppressed	0 turns (0.0%)	1,247 turns (98.42%)
Kinetic Propulsion Energy Preserved	0.0 J	29,493.65 J (29.49 kJ)
Forced Overrides on Node Loss	N/A	7 overrides
Failover Coverage Guarantee	Incomplete / Manual	100.0% Reallocated
EW-CVT Energy Exponent (α)	$\alpha = 0.0$ (Unweighted)	$\alpha = 1.0$ (Proposed)
Jain's Energy Fairness Index ($\mathcal{J}$)	0.9997	0.9991
Fleet Residual Battery Spread (ΔB)	5.08%	8.41%
Fleet Battery Standard Deviation (σ_B)	1.57%	2.84%
Fleet Endurance Safety Margin	+78.46%	+75.13%

In an unconstrained Voronoi implementation, continuous computational ticks and sensor noise would trigger 1,267 trajectory perturbations. The dual-threshold hysteresis loop suppressed 1,247 oscillatory turns (98.42% suppression rate), executing only 20 discrete updates (13 steady updates and 7 failover overrides). This preserved $29,493.65\,\text{J}$ ($29.49\,\text{kJ}$) of kinetic rotor propulsion energy across the fleet.

Furthermore, the EW-CVT energy weighting exponent $\alpha = 1.0$ dynamically balanced operational workloads, factoring residual battery reserves into failover reassignment costs. Surviving UAVs achieved a Jain's Energy Fairness Index of $\mathcal{J} = 0.9991$, maintaining a tight residual battery spread of $\Delta B = 8.41\%$ (min 90.13%, max 98.54%, mean 93.63%) with standard deviation $\sigma_B = 2.84\%$. Crucially, all 9 surviving UAVs concluded search sweeps with a +75.13% endurance safety margin above the mandatory 15% Return-to-Launch safety threshold.

V. Conclusion

This paper presented an integrated, end-to-end autonomous multispectral UAV swarm framework for post-disaster search and rescue. By unifying a 4-channel early-fusion YOLOv8n-P2 detector with an RF physics-informed Delay-Tolerant EdgeBroker, a dual-speed Base Station flight corridor envelope, and an anti-chattering Dual-Threshold Hysteresis EW-CVT controller, the framework successfully overcomes the compounded bottlenecks of small-target aerial perception, multi-kilometer wireless disconnection, and flight path chattering instability. Extensive benchmark experiments across a $16.0\,\text{km}^2$ disaster arena with 10 UAVs demonstrated 100.0% area survey coverage, 71.04% in-flight victim recall (2.15 m localization accuracy, 3.22 m RMSE), 98.42% chattering suppression saving 29.49 kJ of rotor energy, and autonomous in-flight failover recovery. Future work will deploy the architecture onto physical hexarotor testbeds equipped with Jetson Orin edge accelerators and software-defined radio transceivers.

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